# HAIRUL AZHAR BIN ABU BAKAR

**FORWARD-DEPLOYED AI ENGINEER | AI SOLUTIONS ARCHITECT | FOUNDER**

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**PROFILE:** Forward-deployed AI engineer and founder who builds agentic AI and runs it in production. Through Agentic Intelligence Matrix (AiM Intelligence), Singapore, I architect **KatalockAi**, a live multilingual AI workspace for India and Southeast Asia; its **523-lesson Academy**; and **AIM OS**, a governed agent platform. I have shipped live client AI products on KatalockAi. Before AI, I spent 10+ years owning budgets, SLAs and on-site delivery in integrated-resort fit-outs, luxury-hospitality supply and enterprise logistics, so I build AI that operators can trust: verified, human-approved and recoverable.

## LIVE PRODUCTS & CLIENT DELIVERY

**KatalockAi**
www.katalock.net
Agentic workspace that researches, builds, markets and creates media in local languages across India and SEA, with K-Computer verification.

**KatalockAi Academy**
www.katalock.net/academy
20 phases, 523 lessons, 6 career routes; Professor Katalock tutors, reviews and grades student code.

**AIM OS**
aimglobal.io
Governed agent OS: 50+ apps, 108 API endpoints, 27+ agents, QGov policy engine and killswitch.

**Tarot with Shilpa · client AI product**
www.tarotwithshilpa.com
Persona-based AI readings in the user's language, freemium metering, sign-in and hand-off to a human reader.

**Ignis Eco · client website**
www.igniseco.com
Investor site for a plastic-to-fuel venture: crisis, process, plant and team, with a language switcher.

## TECHNICAL SKILLS & ARCHITECTURE STACK

**AI & Agentic Engineering**

Agentic AI · Multi-Agent Orchestration · Cross-Model Verification · RAG · Model Context Protocol (MCP) · Knowledge Distillation · Quantization · Evaluation & Release Gates · Human-in-the-Loop Approval · AI Safety & Red-Teaming · Vector DBs (Pinecone, Chroma)

**Full-Stack & Systems Infrastructure**

Python · TypeScript · React / Next.js · Node.js / Express · FastAPI / Django · PostgreSQL / Supabase · Redis · Docker / Kubernetes · AWS / GCP · Terraform · Stripe

## PROFESSIONAL EXPERIENCE

**AI Platform Architect & Co-Founder | KatalockAi (www.katalock.net)** — **2026 – Present**
*Agentic Intelligence Matrix (AiM Intelligence) · Singapore / India · Remote*

- Own backend architecture, agent orchestration and infrastructure end to end for an agentic AI workspace built by a small team across Singapore and India; co-founder runs on-the-ground operations.
- Built a tiered model stack (katalock-lucy, -tara, -stargazer, -aim) using **knowledge distillation and cost-performance routing**, localised in language, compliance (GST/UPI) and currency for India, Singapore and Indonesia.
- Shipped **K-Computer** cross-model verification with confidence levels, browser automation that pauses for approval before any submit or payment, an evidence-ledger CRM, and a code sandbox that builds, publishes and pushes to GitHub.
- Designed a human-approval rule across the product: no message, payment or post leaves the workspace without a person saying yes. Priced from ₹299/month up to done-for-you agent teams.
- Delivered client AI products: **Tarot with Shilpa** (persona-based readings, freemium metering, human hand-off), the **Ignis Eco** investor site, AI video for property marketing, and SME bundles for clinics, retail, real estate and restaurants.

**Architect & Founder | KatalockAi Academy (www.katalock.net/academy)** — **2026 – Present**
*Global platform · Remote*

- Built a **20-phase, 523-lesson** AI engineering platform with 6 career routes, 6 skill paths and 17 certified capstones; exams issue certificates with verifiable IDs.
- Built **Professor Katalock**, a Socratic AI tutor in every lesson that checks understanding, reviews student code, flags edge cases and grades projects against rubrics.

- Curriculum spans first-principles maths to production agents: LLM serving (vLLM, SGLang, TensorRT-LLM, KV-cache optimisation), quantization (AWQ, GPTQ, FP8, NVFP4), alignment (RLHF, DPO, Constitutional AI), MCP and multi-agent systems.

**Founder & AI Platform Architect** | **Agentic Intelligence Matrix · AIM OS (aimglobal.io)** **Jan 2023 – Present**
*Singapore · Remote*

- Founded a Singapore AI company rooted in **AI safety and red-teaming**: stress-testing models to find where they hallucinate, leak and fail, then building guardrails first.
- Architected AIM OS, an operating system for agentic AI spanning 50+ integrated apps, 108 API endpoints, 27+ autonomous agents and about 80 database tables.
- Designed the **QGov** policy engine, **Sentinel** safety monitor and a founder killswitch; built **Nexus Hub** orchestration with intent routing, pg-boss retries and Yjs CRDT collaboration.

**Co-Founder, Investor Relations & GTM** | **Ignis Eco (www.igniseco.com)** **2026 – Present**
*Tamil Nadu / Bengaluru, India*

- Lead investor relations, seed-stage capital structuring and go-to-market for a plastic-to-fuel pyrolysis venture.

**Logistics Technology Operations (contract)** | **GOGOVAN Singapore (GOGOX)** **Jul 2024 – Sep 2024**
*Singapore*

- Ran enterprise logistics technology for IKEA Singapore: **99.8% on-time delivery** across 5,000+ daily orders, with incident resolution 25% faster.

**Co-Founder & Operations Director** | **Freda'D Pte Ltd / DGCC Investment Pte Ltd** **Apr 2014 – Dec 2024**
*Singapore / Middle East · run concurrently with employment roles*

- Grew a bootstrapped fragrance business from $500 to a profitable international supplier to luxury hotel portfolios in Asia and the Middle East, at **25% YoY growth**.
- Deployed automation and inventory systems in client operations, cutting overhead 30% and lifting engagement 40% at 99.8% on-time delivery.

**Project Director** | **HH Design Pte Ltd / Interior Producer Pte Ltd** **Jan 2010 – Mar 2020**
*Singapore*

- Delivered integrated-resort fit-outs including **Marina Bay Sands Casino** and **Resorts World Sentosa**, owning $500K–$5M budgets and improving timelines by 15%.

## FEATURED PROJECTS & SYSTEMS

**1. KatalockAi Workspace** www.katalock.net
**Next.js, Supabase, distillation, Composio, Docker, Stripe**
Multilingual agent workspace with tiered models, K-Computer verification, gated browser automation and a code sandbox.

**2. KatalockAi Academy** www.katalock.net/academy
**Next.js, Supabase, automated evaluation, rubric grading**
523-lesson platform where an AI tutor reviews code and grades projects; certificates with verifiable IDs.

**3. Tarot with Shilpa** www.tarotwithshilpa.com
**Built with KatalockAi: LLM personas, multilingual UI, auth, metering**
Consumer AI reading app with reader personas, spreads, a monthly free tier and escalation to a human reader.

**4. WorkflowGenius**
**github.com/CybergigSG/WorkflowGenius**
**Python, RAG, Pinecone, React, AWS Lambda**
Enterprise RAG pipeline processing 10,000+ daily transactions at 99.5% uptime, cutting admin workload 40%.

## EDUCATION & CREDENTIALS

- **MSc in Computer Science** — Heriot-Watt University (expected May 2028)
- **Diploma in Architectural Technology** — Singapore Polytechnic (2001–2004)

## PROFESSIONAL CERTIFICATIONS

- **IBM Full Stack Software Developer** (2024) — React, Python, Docker, K8s
- **Google AI & Machine Learning** (2024) — GenAI, data science
- **Google Cybersecurity Professional Certificate** (2024)

## ADDITIONAL INFORMATION

**Languages:** English (native), Malay (native)
**Work authorisation:** Singapore citizen. Open to relocation: USA, UK, EU, UAE, Australia, Canada, India, APAC.
**Availability:** Immediate, or two weeks' notice.